

MOORING PIER AT TROY LOCK AND DAM RTA DESIGN ANALYSIS

ALBANY, NY



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New York District

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SECTION 1 - PROJECT DETAILS

1.1 BACKGROUND

The Troy Lock and Dam mooring pier is located downstream of Troy Lock on the Hudson River. The mooring pier is currently used by public and private ships when preparing to enter and exit the lock. The mooring pier, constructed in 1968, is a diaphragm type cellular structure. It consists of seventeen steel ½ inch sheet pile cellular filled with crushed stone and covered by a concrete deck. The mooring pier has a timber fender system consisting of seven rows of oak rubbing strips that are fastened along the entire length of the pier. There are nine cast iron snubbing posts and eight ladders.

1.2 EXISTING CONDITIONS

Based on two investigations by the NAN Structural design team determined that the current mooring pier cellular structures are in a stable condition and do not require replacement at this time. The determined the necessary repairs and upgrades that inform the scope of this project. The Final Structural Inspection Report has been included as APPENDIX B.

1.3 PROJECT SCOPE

In general terms, the project scope consists of:

1. Timber rubbing fenders replacement with composite material fendering system
2. Fendering system supports and tie plates replacement
3. Ladders and metal grating replacement (as a bid option)
4. Cell walls repair
5. Clearing of vegetation overhanging the mooring pier
6. Concrete decks replacement
7. Analysis of Existing Conditions of the Cell Walls

SECTION 2 - STRUCTURAL

2.1 GENERAL

2.1.1 DESCRIPTION OF THE EXISTING STRUCTURE

Through the original Troy Lock & Dam Mooring Pier (TLDMP) 1968 design documents, the structural team could determine the existing information mentioned in this section. The TLDMP is an approximately 413-foot steel cellular structure. It consists of seventeen steel pile cells constructed with ½ inch thick S-32 sheet piles. The steel cells are filled with crushed stone and covered by a concrete deck.

The existing TDLMP has a timber fender system consisting of seven rows of oak rubbing strips that are fastened along riverside of the hydraulic steel cells along the entire length of the pier. The rubbing strips are either directly connected to the cell or to vertical steel 12BP53 posts that are braced against the cells with 9C20 Channels. The strips are connected to both elements through a bolted connections to a double angle that is welded to the hydraulic cells and the posts. The ends of the Mooring Pier are protected by a steel fender system consisting of seven rows of steel channel rubbing strips that are connected to the steel cells through a set of double angles that are welded to both elements. The steel channel rubbing strips are parallel to the timber rubbing strips.

In addition to supporting the timber fenders, the vertical steel posts and channel struts support a grating that is level to the top of the hydraulic cells. There are additional cantilevered channels that are welded to the hydraulic steel cells which also support the grating. Channel struts located at a lower elevation along the steel posts serve only to brace the post against lateral loads.

There are nine cast iron snubbing posts that are cast into the concrete decks at the top of the cells. There are also eight ladders that reach a landing notched into the steel cell walls and the concrete deck. The locations of the snubbing posts and the ladders are at alternating steel cells. Therefore, the snubbing posts are located at every odd-numbered steel cell while the ladders are at every even-numbered steel cell. For the purposes of the structural design, these are referred to as “Snubbing Post Cells” and “Ladder Cells” respectively.

2.1.2 SUMMARY OF FINDINGS FROM THE NOVEMBER, 2022 INSPECTION

An inspection was conducted in November 2022. A summary of the inspection and its findings, titled “Structural Observations of Mooring Pier - Memorandum”, was issued on January, 2023. The following key deficiencies were found:

- A majority of the timber rubbing strips show significant deterioration, are disconnected from the angles, or are removed completely from the structure.
- The bolted angle connections show missing bolts, corroded torched holes, significant bending, and rusted welds.

- One steel post shows section loss.
- One steel post shows a bent flange.
- The welds in the angle connections show rusting and general corrosion.
- The sheet pile shows section loss in a few landside locations.
- Leaks in the sheet pile indicate a deteriorated joint or steel.
- The sheet pile paint is peeling and the steel show signs of rusting.
- The welded connections at the channel struts and channel cantilevers supporting the grating and the steel posts show significant corrosion.
- Several components such as the ladders and the grating are bent and deformed.

The inspection team found that the existing condition of the mooring pier cannot perform its intended function and must be repaired. While the hydraulic steel cells and the steel posts do not need immediate replacement, the fender system and its connection to the steel structure is in significant disrepair.

2.1.3 STRUCTURAL SCOPE OF REPAIRS

A full replacement is not deemed feasible nor immediately necessary as the hydraulic steel cells and steel posts are mostly in acceptable condition. The scope of the repairs are as follows:

- The design and construction of a new structural frame on the west side of the TLDMP. This structural frame will be supported by the existing TLDMP hydraulic steel cells and 12BP53 vertical steel posts. The structural frame's primary purpose will be to support a new marine fender product.
- The analysis and replacement of steel channels that support the steel post and the walkway grating.
- The repair of leaking hydraulic steel cell walls observed on the downstream cell.
- The demolition and replacement of existing concrete decks at the ladders with significant settlement.
- The repair or, if needed, replacement of the ladders and walkway grating.
- The analysis of the current capacity of the 12BP53 posts considering the work mentioned in this list and the deficiencies found in the field.
- The analysis of the current capacity of the hydraulic steel cell wall considering the work mentioned in this list and the deficiencies found in the field. This analysis will be for internal use only and is intended to be referenced for future projects.

2.2 STRUCTURAL CODES AND REFERENCES

The structural design and analysis of the TLDMP fender system and existing conditions are in accordance with the following codes and standards where applicable:

- ASCE/SEI 7-16: Minimum Design Loads and Associated Criteria for Buildings and Other Structures
- UFC 3-301-01 with Change 1 (October 2, 2023) Structural Engineering
- International Building Code (IBC) (2021)
- International Existing Building Code (IEBC) (2021)
- EM 1110-2-2107 Design of Hydraulic Steel Structures (2022) *
- EM 1110-2-2502 Flood Walls and other Hydraulic Retaining Walls (2022)
- EM 1110-2-2607 Planning and Design of Navigational Dams (1995)
- EM 1110-2-3402 Barge Impact Forces for Hydraulic Structures (2022)
- AISC Steel Construction Manual (4th Edition, 1941)
- AISC Steel Construction Manual (15th Edition, 2017)
- AISC Design Guide 9 Torsional Analysis of Structural Steel Members (2003)
- AASHTO LRFD Bridge Design Specification (9th Edition, 2020)
- AASHTO LRFD Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals (2015 Edition)
- USS Hot-Rolled Steel, Shapes and Plates (1963)
- 1968 Troy Lock & Dam Mooring Pier Contract Documents

* The TLDMP is not considered a “Hydraulic Steel Structure” since its primary purpose is not to retain water. However, EM 1110-2-2107 is still used as a reference for determining design loads and details since the TLDMP is a steel structure submerged in water.

2.3 DESIGN LOADS

Loads shall be in accordance with the codes mentioned in Section 3. Due to the low-risk of loss-of-life in the event of a failure, the structural loading is determined based on Risk Category I. The site class is also assumed to be B based on the borings and design of the mooring pier originally done in 1968, which stated that the TLDMP is founded on solid and sound rock anywhere between 9 feet and 18 feet below the riverbed. When using the online WDBG tool based on UFC 3-301-01, the design load conditions were determined as follows at the nearby Watervliet Arsenal in Albany, NY:

Latitude / Longitude					42.7204539, -73.71343430000002				
WIND SPEED (MPH)					WIND SPEED (KM/H)				
RISK CATEGORY					RISK CATEGORY				
I	II	III	IV	V	I	II	III	IV	V
102	110	118	123	146	164	177	190	198	235
SNOW LOADING									
GROUND SNOW (PSF)		FROST PENETRATION (IN)			GROUND SNOW (KPA)		FROST PENETRATION (MM)		
40		82			1.92		2083		
SEISMIC DATA (SITE CLASS B)									
PGA (%G)		S _s (%G)		S ₁ (%G)		S _{MS} (%G)		S _{M1} (%G)	
10.6		19.8		6		17.8		4.8	

Figure 1: Design Wind, Snow, and Seismic Considerations for Watervliet Arsenal

2.3.1 DEAD LOAD (D)

The following self-weights were assumed in the calculation of Dead Loads as per EM 1110-2-2107:

- Composite Fender Material: 70 lb/ft³
- Concrete: 150 lb/ft³
- Gravel Fill: 140 lb/ft³
- S-32 Steel Sheet pile: 32 lb/ft²
- Steel Grating: 25 lb/ft²
- Steel: 490 lb/ft³
- Water (Fresh): 62.4 lb/ft³

All values were either taken from ASCE 7-16 or the AASHTO LRFD Bridge Design Specification. The exceptions are for the S-32 steel sheet pile (USS Hot-Rolled Steel Shapes and Plates, MP102 Equivalent Shape, 1963) and the composite fender material (Maximum According to Various Product Catalogs).

The assumptions made for this fendering system when determining its load was:

- It doesn't involve pneumatic elements
- The outside face of the system is maximum 2'-0" from the outer flange of the steel wide flange support beam.

2.3.2 GRAVITY LOADS (G or G_{ICE} for icy conditions)

The following self weights were assumed in the calculation of Gravity Loads as per EM 1110-2-2107:

- Compacted Soil: 120 lb/ft³
- Soil at Channel bottom (saturated): 125 lb/ft³
- Soil at Channel bottom (submerged): 62.5 lb/ft³
- Ice: 56 lb/ft³

All values were either taken from ASCE 7-16, the AASHTO LRFD Bridge Design Specification, or the 1968 Mooring Pier Design Memorandum. The heaviest soil is conservatively assumed to fill in the empty space on the topside of the web in the wide flange beam.

The Weight of the Atmospheric Ice is calculated as per ASCE 7-16. The space between the fender support beams are also conservatively assumed to be filled with ice. And the ice gravity load is assumed to only occur in conjunction with Wind-on-Ice (See Section 2.3.6), Ice Impact (See Section 2.3.8.1), and Thermally-Expanding Ice Load Cases (See Section 2.3.9).

2.3.3 LIVE LOAD (L)

A 100 psf live load is considered for the analysis and repair of the steel post, channel shapes, and steel grating (EM 1110-2-2107 4.2.11). This live load is expected to follow a load path from the application on the steel grating, through the channel struts, to the vertical steel posts.

2.3.4 SNOW LOAD (S)

Ground Snow Load = 40 psf per WBDG Structural Load Tool for UFC 3-301-01.

2.3.5 FROST PENETRATION

82 inches per WBDG Structural Load Tool for UFC 3-301-01.

2.3.6 WIND LOAD (W or W_{ICE} for icy conditions or W_{SERV} for Service Load Comb.)

Determined in accordance with ASCE 7-16 using a Basic Wind Speed of 102 MPH per WBDG Structural Load Tool for UFC 3-301-01 for Risk Category I using exposure category “D.”

In accordance with ASCE 7-16, a wind speed of 50 mph was used for cases where the structure is covered in ice. Also, in accordance with ASCE 7-16 and AASHTO LRFD Specifications for Structural Supports for Highway Signs, Luminaires, and Traffic Signals (2015 Edition), the wind speed for service load combinations of 75 mph was based on a mean recurrence interval (MRI) of 10 years. This MRI is typically chosen for sign and luminaire structures, and the flexibility of the fender system and its structural supports can allow us to treat this structural component as a sign structure.

2.3.7 FLOOD LOAD

2.3.7.1 *Hydrostatic Load (H_s)*

While the wind (See Section 2.3.6) and wave (See Section 2.3.7.3) loading assumes a closed-wall type fendering system for maximum loading, water is expected to move freely to and from either side of the fendering system. Therefore, the net lateral loading due to hydrostatic loads is assumed to be negligible.

The buoyant forces of a fully-submerged fender support system was considered. However, the buoyant forces were calculated to not overcome the permanent dead loads in Section 2.3.1. Therefore, these forces were assumed to be negligible as well.

2.3.7.2 *Hydrodynamic Load (H_d)*

As per EM 1110-2-2107 E.2.8, the hydrodynamic loads were estimated by applying a head of 1.25 feet. This load is assumed to be applied either towards or away from the mooring pier.

2.3.7.3 *Wave Load (H_w)*

Without a detailed hydrodynamic study, the breaking wave pressure equations in ASCE 7-16 Section 5.4 were used to estimate the wave loading on the structure. As with the Wind Loads (See Section 2.3.6), the wave loads were conservatively assumed to act on a solid closed wall fender system. Wave loads are also assumed to act perpendicular to the mooring pier. This load is assumed to only occur in the direction towards the mooring pier.

2.3.8 IMPACT LOAD

2.3.8.1 *Debris and Ice Impact Loads (IM)*

The debris impact load was estimated by using the provisions in ASCE 7-16 C5.4.5.

As per EM 1110-2-2502 6.12.3, the impact force due to floating ice is represented as a 500 plf lateral loading. The EM states that this is the maximum impact load that can be expected for structures that are parallel to the water flow. This load is assumed to only occur in the direction towards the mooring pier.

2.3.8.2 *Barge Impact Loads (BI)*

The maximum barge size used in the 1968 design was 5600 tons. The Albany Field Office confirmed with that the size of the maximum barge has not increased since 1968. This tonnage and EM 1110-2-3402 (Barge Impact Forces for Hydraulic Structures) is used to determine barge impact loads for the mooring pier fendering system.

The design for the Troy Lock & Dam Miter Gate (TLDMG) conducted in 2016 has calculated the unsymmetric and symmetric barge impact force based on a 5600-ton barge. The design team in 2016 has determined this barge impact force to be conservative for the Troy Lock & Dam location but acceptable for use. These forces were also used for these calculations.

EM 1110-2-3402 Table D.3 provides preliminary impact angles for approach conditions to navigation lock walls. Because the mooring pier is largely parallel to the Hudson River, these angles were used to determine the impact force both perpendicular and parallel to the fender support system. While impact angles were provided for usual, unusual, and extreme cases, only the extreme case was analyzed as per EM 1110-2-2107 Table 4.1.

Because of the large size of the 5600-ton barge, this load is assumed to be distributed by tributary area to all fender support beams. Conservatively, this load is assumed to be applied in conjunction with basic wind and serviceability wind (see section 2.3.6) load cases. This load is assumed to only occur in the direction towards the mooring pier.

2.3.9 ICE (THERMAL EXPANSION) (IX)

As per EM 1110-2-2107 4.2.7, the force due to thermally-expanding ice is represented as a 5 klf lateral loading. This value is also confirmed in the TLDMG design. Conservatively, this load is applied perpendicularly to one fender support beam. This load is only assumed to be applied in the direction away from the mooring pier. The thermal expansion of ice is a significant load in confined spaces such as locks, but the mooring pier is in the river with no other structures nearby to restrict the expansion of ice. The only potential confined space is between the steel fender support beams and the mooring pier cell walls. Therefore, an expanding ice sheet will only affect the structure in one direction.

2.3.10 SELF-STRAINING LOADS (T)

Self-Straining Loads in the Mooring Pier is assumed to only come from contraction / expansion due to temperature changes. The steel temperature expansion coefficient is 6.50×10^{-6} in/in/°F as per the AISC Steel Construction Manual. Because this is a cold climate, the temperature changes defined in AASHTO LRFD Bridge Design Table 3.12.2.1-1 lead to a significant expansion in the steel fender supports. Therefore, the steel fender support beams shall be connected at the hydraulic cells with slotted bolt holes to allow this expansion and prevent excess axial forces from developing in the beam.

2.3.11 OPERATIONAL LOADS (Q) & FRICTION LOADS (F)

Because the mooring pier doesn't have any operational equipment, any load cases regarding operational equipment and friction from operational equipment were not considered.

2.3.12 SEISMIC LOAD (EQ)

The buildings are classified as Risk Category I with a Seismic Importance Factor, I_E , of 1.00. There is no geotechnical report that was recently prepared, nor is there one being prepared as part of the scope. From the borings that were taken for the original Mooring Pier design conducted in 1968 and the characterizes the site soil class as "B," and the seismic design category as "A."

Spectral Response Acceleration Parameters

$PGA = 0.106g$ (WBDG Structural Load Tool for UFC 3-301-01)

$S_S = 0.198g$ (WBDG Structural Load Tool for UFC 3-301-01)

$S_1 = 0.060g$ (WBDG Structural Load Tool for UFC 3-301-01)

Adjusted MCE_R Spectral Response Accelerations

$S_{MS} = 0.178g$ (WBDG Structural Load Tool for UFC 3-301-01)

$S_{M1} = 0.048g$ (WBDG Structural Load Tool for UFC 3-301-01)

Design Spectral Response Accelerations

$S_{DS} = 2/3 \times S_{MS} = 0.119g$

$S_{D1} = 2/3 \times S_{M1} = 0.032g$

Seismic Design Category:

$S_{DS} = \text{Category A}$ (ASCE 7-16, Table 11.6-1)

$S_{D1} = \text{Category A}$ (ASCE 7-16a, Table 11.6-2)

2.3.13 LOAD COMBINATIONS

After calculating the load cases in the previous sections, the load combinations in the following subsections are determined to control because of the high effects of barge impact and thermal expansion of ice loads. These load combinations are determined with the guidance of EM 1110-2-2107 Table 4.1, Section 3.3, and Section 4.

- Load Combination 1/2 (LC 1 & LC 2):

$$\gamma_{pr} * D + \gamma_p * G_{ICE}$$

- Load Combination 3/4 (LC 3 & LC 4):

$$\gamma_p * (D + G_{ICE}) + \gamma_{pr} * IX + \gamma_c * H_d$$

- Load Combination 5/6 (LC 5 & LC 6):

$$\gamma_p * (D + G) + \gamma_{pr} * BI + \gamma_c * H_d$$

- Load Combination 7/8 (LC 7 & LC 8): (Wind Companion Load Only)

$$\gamma_p * (D + G) + \gamma_c * W$$

- Load Combination 9/10 (LC 9 & LC 10) (Hydrodynamic Companion Only)

$$\gamma_p * (D + G) + \gamma_c * H_d$$

Load combinations 7 through 10 only serve to analyze the post as the barge impact and thermal expansion of ice forces will only apply load to the post through a single or a few fender support beams. Therefore, the other fender support beams will be transferring companion and permanent loads only to the vertical posts.

The other load combinations were considered:

- Load combinations LC S 1 through LC S 10
These loads are to determine the service loads on the structure. They are identical to LC 1 through LC 10 except all the load factors are 1.00 and the wind load used is W_{SERV} instead of W .
- Load combinations LC F 1 through LC F 10
These loads are to determine stresses in the steel components for comparison with their finite fatigue life stress threshold. They are identical to LC 1 through LC 10 except all the load factors are 1.00.
- Load combinations LC N 1 through LC N 10
These loads are to determine stresses in the steel components for comparison with their infinite fatigue life stress threshold. They are identical to LC 1 through LC 10 except all the load factors are 2.00.

2.4 MATERIALS OF CONSTRUCTION

Materials of construction for this facility are as follows:

CONCRETE

Cast-in-place concrete

- | | |
|---------------|--|
| Normal-weight | - 4500 psi (all concrete unless noted otherwise) |
| Reinforcement | - ASTM A615, Grade 60 |

METALS

Structural steel

- | | |
|-------------------------|-------------------------|
| Rolled Steel & Plates | - ASTM A709, Grade 50 |
| Bolts, nuts and washers | - ASTM F3125, Type A325 |
| Welding electrodes | - E70XX |

2.5 STRUCTURE DISCUSSION

2.5.1 STRUCTURAL DESIGN APPROACH

At the time of design, it is currently unknown what fendering product will be used at the mooring pier. USACE designs and solicitations are prohibited from specifying brand names (or equal), or referencing proprietary specifications or standards, without express justification and approval. Therefore, the approach is to design a structural support system that can connect to wide variety of fendering systems and to check the capacity of all structural components based on applied forces and the assumed fender self-weight listed in Section 2.3.

The fenders shall be a composite plastic marine timber product selected by the contractor based on the self-weight assumptions listed in Section 2.3.1 and required impact forces shown on the project drawings. If the self-weight of the selected fender product is heavier than what was assumed during design, then the structure will have to be redesigned.

2.5.2 PROPOSED STRUCTURE DESCRIPTION

This structural support system consists of a series of wide-flange steel beams spanning between the existing hydraulic steel cells and the steel posts. These steel beams shall be oriented so that the stiffest section will be resisting lateral forces due to wind, hydraulic loading, and impact forces. These steel beams will not use the existing angles and will be connected through a direct bolt connection to the existing steel post and a steel bracket that is to be field welded to the side of the hydraulic steel cell.

To ensure that the existing structure can handle the applied loads, the channels used to support the steel post against lateral loading and the grating at the top of the cell will be replaced. The new channels will be field welded to the hydraulic steel cell and the steel post. A new set of channel struts will be field welded above the water line to avoid the need for dewatering operations.

2.5.3 DESIGN CAPACITIES

2.5.3.1 *Strength Limit State*

In order to obtain the proposed and existing-to-remain steel shapes of the fendering support system, the following references were used:

- AISC Steel Construction Manual (15th Edition) Table 1-1:
Proposed steel wide flange fender support beam properties.
- AISC Steel Construction Manual (4th Edition) H-Bearing Piles:
Existing-to-remain steel post properties
- USS Hot Rolled Steel Shapes and Plates (1963 Edition)
Existing-to-remain steel sheeting properties. The USS designation MP102 matches the USACE designation S-32 Sheet Pile. This specification also confirms the section properties listed in the 1968 Plan Set.

The capacities of the steel shapes, welded connections, and bolted connections will be analyzed as per the 15th Edition of the AISC Steel Construction Manual, ANSI/AISC 360-16 Specification for Structural Steel Buildings within the 15th Edition of the AISC Steel Construction Manual.

2.5.3.2 *Fatigue Limit State*

The fatigue life of these structural components will be determined based on these stress reversals and the fatigue detail categories listed in AASHTO LRFD Bridge Design Specification Table 6.6.1.2.3-1. The following fatigue categories are considered:

- Fatigue Detail Category C:
 - Base Metal at Copes
 - Weld Access Holes
 - Web Stiffeners
 - Welded connections.
- Fatigue Detail Category D:
 - Base Metal at Bolt Connections

2.5.3.3 *Serviceability Limit State*

While the greatest loads will be lateral, the deflection limits due to “vertical loading” in ASCE 7-16 Appendix CC.2.1 will be considered. Since this structural system is not primarily used for human occupants, the lowest deflection limit $L/200$ will be considered.

Furthermore, the fender support beams fundamental frequency of vibration (ASCE 7-16 Appendix CC.2.3) will be checked against the frequency due to the applied loads.

SECTION 3 - CIVIL

3.1 GENERAL

Troy Lock and Dam is located on the Hudson River within the city limits of Troy, Rensselaer County, NY. The project lies approximately 160 miles north of New York Harbor, approximately 8.3 miles north of Albany, and one mile downstream of the confluence of the Fifth Branch of the Mohawk River with the Hudson. The Troy Lock and Dam is the first lock upstream of the harbor and serves vessels passing from the Hudson River into the Great Lakes by way of the New York State Canal System.

The Troy Lock and Dam sits at the northern end of the Hudson River tidal estuary.

Congress authorized the construction of the Federal Lock and Dam in the River and Harbor Act of 25 June 1910 as part of the Hudson River, New York navigation improvements contained in House Document 719, 61st Congress, 2nd Session. The Permanent Appropriations Repeal Act of 26 June 1934 included operation and maintenance of the dam in the Hudson River Project effective 1 July 1935. Water is managed at Troy Lock and Dam for navigation and recreation.

The current downstream mooring wall was constructed in 1969. The wall is located approximately 700 feet south of the downstream miter gate. The wall is 414 feet long constructed with steel sheetpile end cells and intermediate diaphragms. The wall has a 1 – 2 foot thick reinforced concrete cap and timber fenders.

3.2 SITE CLEARING/DEMOLITION

Site clearing will be limited to the removal of overhanging tree branches and vegetation growth on the mooring pier. Contractor will be pruning any tree branches that overhang beyond the landside (eastern) edge of the pier. Contractor shall coordinate with the Albany Field Office (AFO) and COR for any tree removal. AFO believes that there will be only one (multi-stemmed) tree to be removed and five trees to be pruned. AFO has record of concurrence from the City of Troy for any tree removal that is required.

There is no demolition as part of this project.

3.3 SITE EARTHWORK

Proposed vegetation removal and structural repairs will require any earthwork activities.

3.4 SITE UTILITIES

Proposed vegetation removal and structural repairs will not impact or be impaired by utility locations.

3.5 SITE ACCESS/STAGING

The contractor will stage and park at the Albany field office, exact location to be

coordinated with the AFO. Contractor will moor their own craft at the upstream mooring wall, adjacent to the boat launch. The contractor will coordinate with the Albany Field Office and COR as to the staging limits and which entrance to the site to use.

3.6 EROSION + SEDIMENT CONTROL

Contractor will be responsible for meeting all the requirements and standards set in the New York State Standards and Specifications for Erosion and Sediment Control (Nov. 2016).